Algorithm 1 f-DAgger

Input: expert π^*, clean demonstrations $\mathcal{D}_{\text{clean}}$, number of DAgger iterations N
Output: best policy $\hat{\pi}_i$ on validation

Phase 1: expert demonstrations with OU noise

1: $\mathcal{D} \leftarrow \mathcal{D}_{\text{clean}}$
2: **for** each training scenario, twice **do**
3: $z \leftarrow 0$; draw a noise scale $\alpha \sim \mathcal{U}[0.25, 1]$
4: **for** each time step t **do**
5: $a \leftarrow \pi^*(s_t)$; record (s_t, a) ▷ clean expert label
6: $z \leftarrow \rho z + \sqrt{1-\rho^2}\,\alpha\sigma\xi,\ \xi \sim \mathcal{N}(0, I)$ ▷ OU noise
7: execute $\text{clip}(a + m_t z)$ ▷ m_t: noise gate, Table 1
8: add the episode to $\mathcal{D}$ if the expert holds the target
9: train $\hat{\pi}_0$ on $\mathcal{D}$

Phase 2: DAgger

10: **for** $i = 1, \ldots, N$ **do**
11: run $\hat{\pi}_{i-1}$ on the training and new development scenarios
12: get dataset $\mathcal{D}_i = \{(s, \pi^*(s))\}$ of visited states and expert actions
13: aggregate datasets: $\mathcal{D} \leftarrow \mathcal{D} \cup \mathcal{D}_i$
14: train $\hat{\pi}_i$ on $\mathcal{D}$
15: **return** the $\hat{\pi}_i$ $(i \geq 1)$ with the highest validation success rate

OU noise. Line 6 is the exact discrete-time Ornstein–Uhlenbeck process [1], with $\rho = e^{-\Delta t/\tau}$. Each channel of z has standard deviation $\alpha\sigma$ and correlation time τ. As in DART [2], noise only changes the executed action; the recorded label is the clean expert action.

Noise gate. m_t switches noise off per channel group as the episode progresses (Table 1). Wheel noise stays off once the expert starts reaching. The target counts as reached when the end-effector error is below 10 mm after the first 8 s.

Table 1: Noise gate m_t.

Episode stage	Arm noise	Wheel noise
before the expert starts reaching	on	on
reaching, until 1.5 s after the target is first reached	on	off
from 1.5 s after the target is first reached	off	off

DAgger. Lines 11–14 follow Algorithm 3.1 of [3] with $\beta_i = 0$: the learner acts alone, and Phase 1 replaces DAgger's expert-only first iteration.

Details. An episode ends 3 s after the target is first reached, or at 40 s. The expert *holds the target* if the error stays below 10 mm for the last second of these 3 s; validation success uses the same test. DAgger keeps failed tasks and discards numerically failed rollouts. Each training run starts from a new network and uses the BC+J objective. Commands are clipped to the robot's limits, and a is clipped before noise is added. Ties in validation go to the later iteration.

Defaults. $\Delta t = 1$ ms, $\tau = 0.2$ s, $\sigma = 0.75$ N m (arm) and 0.15 rad/s (wheels), $N = 3$.

References.

[1] G. E. Uhlenbeck and L. S. Ornstein, "On the theory of the Brownian motion," *Physical Review*, vol. 36, no. 5, pp. 823–841, 1930.

[2] M. Laskey, J. Lee, R. Fox, A. Dragan, and K. Goldberg, "DART: Noise injection for robust imitation learning," in *Proc. CoRL*, PMLR, vol. 78, 2017, pp. 143–156.

[3] S. Ross, G. J. Gordon, and J. A. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. AISTATS*, JMLR W&CP, vol. 15, 2011, pp. 627–635.